

CONTINUOUS BACKBONE — SINGLE SPATIAL PASS

Flattened Event Tensor

Channel-wise stack of $T = 10$ bins

[B, 20, 180, 320]

STAGE 1

Initial Stem

Stride-2 Conv + BN + ReLU

[B, 64, 90, 160]

STAGE 2A

Backbone Alpha — Block 1

ResBlock: $64 \rightarrow 128$, stride 2

[B, 128, 45, 80]

STAGE 2B

Backbone Alpha — Block 2

ResBlock: $128 \rightarrow 256$, stride 2

[B, 256, 22, 40]

STAGE 3

Attention Bridge

Squeeze-and-Excitation

[B, 256, 22, 40]

STAGE 4

High-Res Backbone

ResBlock $\times 2$ — stride 1

[B, 512, 22, 40]

HEAD

Channel Projection

1×1 Conv: $512 \rightarrow 128$

[B, 128, 22, 40]

HEAD

Dual Adaptive Pool

AvgPool + MaxPool @ 8×8

[B, 128, 8, 8]

HEAD

Multi-Box Regression Head

Conv + Linear $\rightarrow 5$ boxes

[B, 5, 5]

Bounding-Box Predictions

$5 \times \{x, y, w, h, c\}$